

Q1. What is Machine Learning? Explain supervised and unsupervised learning.

Ans - Machine Learning is a branch of artificial intelligence where systems learn patterns from data and improve performance without being explicitly programmed for each task. Supervised learning uses labeled data, meaning input-output pairs are known during training. The model learns a mapping function and predicts outputs for unseen inputs. Examples: spam detection, house price prediction. Unsupervised learning uses unlabeled data. The model finds hidden structures or patterns, such as clusters or associations. Examples: customer segmentation, anomaly detection. Key difference: supervised learning predicts known target labels, while unsupervised learning discovers structure without target labels.

Q2. What is overfitting? Mention two methods to reduce it.

Ans - Overfitting occurs when a model learns training data too closely, including noise, and performs poorly on new/unseen data.

Two methods to reduce overfitting:

- 1) Regularization (L1/L2): adds penalty to large weights, reducing model complexity.
- 2) Cross-validation / early stopping: monitor validation performance and stop training before the model starts memorizing noise.

Other methods include dropout, data augmentation, and pruning.